Internal monoidal structures and Day Convolution product

in a monoidal 2-category

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February 17, 2023

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Overview

Monoidal structure $(\mathcal{C}, \otimes_\mathcal{C})$ on a $\mathcal{V}$-enriched category $\overset{Day}{\Rightarrow}$ Its $\mathcal{V}$-category of $\mathcal{V}$-functors $[\mathcal{C}, \mathcal{V}]_\mathcal{V}$ inherits a monoidal structure

How does it works ?

$F, G : \mathcal{C} \to \mathcal{V}$
$F \otimes_{Day} G : \mathcal{C} \to \mathcal{V}$

$$\begin{tikzcd} \mathcal{C} \times \mathcal{C} \arrow[r, "F\bar{\otimes}G"] \arrow[d, "\otimes_\mathcal{C}"'] \arrow[r, phantom, shift right=3, "\eta \Downarrow"] & \mathcal{V} \\ \mathcal{C} \arrow[ur, "Lan_{\otimes_\mathcal{C}} F\bar{\otimes}G"'] & \end{tikzcd}$$

$$\otimes_{Day} : [\mathcal{C}, \mathcal{V}]_\mathcal{V} \times [\mathcal{C}, \mathcal{V}]_\mathcal{V} \to [\mathcal{C}, \mathcal{V}]_\mathcal{V}$$

Is it possible to generalize Day convolution product ?

Precisely: Which properties does the 2-categories $\mathrm{C{\small AT}}$ and $\mathrm{C{\small AT}}_{\mathcal{V}}$ share so that Day convolution is possible ?

What do we need ?

$$
\begin{tikzcd}
\mathcal{C} \times \mathcal{C} \arrow[r, "F\bar{\otimes}G"] \arrow[d, "\otimes_{\mathcal{C}}"'] \arrow[dr, phantom, "\Downarrow"] & \mathcal{V} \\
\mathcal{C} \arrow[ur, "Lan_{\otimes_{\mathcal{C}}} F\bar{\otimes}G"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
{[\mathcal{C}, \mathcal{V}]_{\mathcal{V}} \times [\mathcal{C}, \mathcal{V}]_{\mathcal{V}}} \arrow[r, "\otimes_{Day}"] \arrow[d] & {[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}} \\
{[\mathcal{C} \times \mathcal{C}, \mathcal{V} \times \mathcal{V}]_{\mathcal{V}}} \arrow[d] & \\
{[\mathcal{C} \times \mathcal{C}, \mathcal{V}]_{\mathcal{V}}} \arrow[uur, "Lan_{\otimes_{\mathcal{C}}}"'] &
\end{tikzcd}
$$

1. A 2-category Ω
2. An object of morphisms, thus a closed structure
3. Kan extensions
4. A monoidal structure $(\Omega, \otimes_\Omega)$ on Ω and monoids with respect to this structure
5. An external product

Formal category theory

Idea

Gray in *Formal category theory*:
" The basic idea is that the category of small categories, *Cat*, is a 2-category with properties and that one should attempt to identify those properties that enable one to do the “structural parts of category theory”."

Extending categorical constructions in a 2-category that looks like Cat, for instance:

- Adjoint morphisms
- Kan extensions

Let Ω be a 2-category.

Local left Kan extension

When it exists: $(\alpha_F, Lan_\mu F)$ universal among such couples in Ω.

$$\begin{tikzcd} \mathcal{C}_2 \arrow[r, "F"] \arrow[d, "\mu"'] & \mathcal{V} \\ \mathcal{C}_1 \arrow[ur, "Lan_\mu F"'] \arrow[u, phantom, "\alpha_F \Downarrow"] & \end{tikzcd} .$$

Global left Kan extension

When it exists, left adjoint functor Lan_μ of the *functor* induced by a morphism $\mu : \mathcal{C}_2 \to \mathcal{C}_1$ in Ω:

$$Lan_\mu : \qquad \Omega(\mathcal{C}_2, \mathcal{V}) \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \Omega(\mathcal{C}_1, \mathcal{V}) \qquad : \Omega(\mu, \mathcal{V})$$

Here the notion of adjointness inside a 2-category is not needed to define Kan extensions.

Theorem
The global left Kan extension along a morphism μ exists if and only if the local left Kan extension along it exists for all morphism. In this case, if it exists along any morphism $\mathcal{C}_2 \to \mathcal{C}_1$, we obtain a functor

$$Lan : \Omega(\mathcal{C}_2, \mathcal{C}_1)^{op} \longrightarrow [\Omega(\mathcal{C}_2, \mathcal{V}), \Omega(\mathcal{C}_1, \mathcal{V})] .$$

Let $(\Omega, \times, [-,-]_\Omega)$ be a cartesian closed 2-category, $\mu : \mathcal{C}_2 \to \mathcal{C}_1$ be a morphism in Ω and $\mathcal{D} \in \Omega_0$ be an object.

Definition
The internal left Kan extension along μ is defined, when it exists, as the left adjoint in Ω of $[\mu, \mathcal{D}]_\Omega$:

$$Lan^\Omega_\mu : \quad [\mathcal{C}_2, \mathcal{D}]_\Omega \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} [\mathcal{C}_1, \mathcal{D}]_\Omega \quad : [\mu, \mathcal{D}]_\Omega \; .$$

Proposition
If the internal left Kan extension exists along any morphism $\mathcal{C}_2 \to \mathcal{C}_1$, it extends to a functor

$$Lan^\Omega : \Omega(\mathcal{C}_2, \mathcal{C}_1)^{op} \to \Omega\left([\mathcal{C}_2, \mathcal{D}]_\Omega, [\mathcal{C}_1, \mathcal{D}]_\Omega\right).$$

Since Ω is a 2-category, we have a 2-functor

$$\Omega(-,-) : \Omega^{op} \times \Omega \to \textsc{Cat}.$$

It is a morphism of closed 2-categories: we have a functor

$$\eta : \Omega\left([\mathcal{C}_2, \mathcal{D}]_\Omega, [\mathcal{C}_1, \mathcal{D}]_\Omega\right) \to [\Omega(\mathcal{C}_2, \mathcal{D}), \Omega(\mathcal{C}_1, \mathcal{D})]$$

and obtain a commutative triangle

$$\begin{tikzcd}
\Omega(\mathcal{C}_2, \mathcal{C}_1)^{op} \arrow[r, "Lan^\Omega"] \arrow[dr, "Lan"'] & \Omega\left([\mathcal{C}_2, \mathcal{D}]_\Omega, [\mathcal{C}_1, \mathcal{D}]_\Omega\right) \arrow[d, "\eta"] \\
& {[\Omega(\mathcal{C}_2, \mathcal{D}), \Omega(\mathcal{C}_1, \mathcal{D})]}
\end{tikzcd}.$$

Internal structures in a monoidal 2-category

Let the 2-category Ω be equipped with a 2-functor

$$\otimes_\Omega : \Omega \times \Omega \to \Omega$$

associative up to an associator, and a unit $1_\Omega \in \Omega_0$ which is unital up to a unitor.

Examples

- $(\mathrm{CAT}, \times)$
- $(\mathrm{CAT}_{\mathcal{V}}, \times)$
- $(Mon\mathrm{CAT}, \otimes)$, where $(\Omega_1, \otimes_{\Omega_1}) \otimes (\Omega_2, \otimes_{\Omega_2}) = (\Omega_1 \times \Omega_2, \otimes_{\Omega_1^2})$, with

$$\otimes_{\Omega_1^2} : \Omega_1 \times \Omega_2 \times \Omega_1 \times \Omega_2 \xrightarrow{\cong} \Omega_1 \times \Omega_1 \times \Omega_2 \times \Omega_2 \xrightarrow{\otimes_{\Omega_1} \times \otimes_{\Omega_2}} \Omega_1 \times \Omega_2$$

- $(\mathrm{CAT}^{\mathbb{N}}, \circ)$, where $\mathcal{C} \circ \mathcal{D}(n) = \coprod_{n_1 + \ldots + n_r = n} \mathcal{C}(r) \times \prod_{i=1}^{r} \mathcal{D}(n_i)$

Let $(\Omega, \otimes_\Omega)$ be a monoidal 2-category.

Pseudomonoid internal to $(\Omega, \otimes_\Omega)$

- An object $\mathcal{C} \in \Omega_0$,
- Unit and composition morphisms

$$1_{\mathcal{C}} : 1_\Omega \to \mathcal{C}, \quad \mu_{\mathcal{C}} : \mathcal{C} \otimes_\Omega \mathcal{C} \to \mathcal{C} \qquad \in \Omega_1,$$

- Unitors and associators 2-morphisms

$$\alpha_{\mathcal{C}} : \mu_{\mathcal{C}} \circ (\mu_{\mathcal{C}} \otimes_\Omega id_{\mathcal{C}}) \Rightarrow \mu_{\mathcal{C}} \circ (id_{\mathcal{C}} \otimes_\Omega \mu_{\mathcal{C}})$$

  verifying coherence conditions.

Pseudomonoidal morphism between pseudomonoids

- morphism $F : \mathcal{C} \to D$ in Ω

- with 2-morphisms in Ω

$$\begin{tikzcd} 1_\Omega \arrow[r, "1_\mathcal{C}"] \arrow[dr, "1_\mathcal{D}"'] & \mathcal{C} \arrow[d, "F"] \\ & \mathcal{D} \arrow[u, phantom, shift right=3, "\Uparrow e_F"] \end{tikzcd} \qquad \begin{tikzcd} \mathcal{C} \otimes_\Omega \mathcal{C} \arrow[r, "\mu_\mathcal{C}"] \arrow[d, "F \otimes_\Omega F"'] & \mathcal{C} \arrow[d, "F"] \\ \mathcal{D} \otimes_\Omega \mathcal{D} \arrow[r, "\mu_\mathcal{D}"'] \arrow[ur, Rightarrow, "m_F"'] & \mathcal{D} \end{tikzcd}$$

verifying unit and associativity conditions.

Together with 2-morphisms $F \Rightarrow G$ in Ω strictly preserving structural morphisms of F and G, we obtain a 2-category

$$PsMon(\Omega, \otimes_\Omega).$$

- $\mathit{Mon}\textsc{Cat}$ = the 2-category of monoidal categories with lax monoidal morphisms and natural transformations preserving their structure, then

$$\mathit{PsMon}(\textsc{Cat}, \times) \cong \mathit{Mon}\textsc{Cat}$$

- $\mathit{PsMon}(\textsc{Cat}_{\mathcal{V}}, \times) \cong \mathit{Mon}\textsc{Cat}_{\mathcal{V}}$
- $\mathit{PsMon}(\mathit{Mon}\textsc{Cat}, \otimes) \cong \mathit{Mon}^2\textsc{Cat}$
- $\mathit{PsMon}(\textsc{Cat}^{\mathbb{N}}, \circ) \cong \mathit{PsOp}_{\textsc{Cat}}$
- $\mathit{Mon}\textsc{Cat}_2$ = 2-category of monoidal 2-categories, pseudomonoidal 2-functors.

$$\mathit{Mon}\textsc{Cat}_2 \cong \mathit{PsMon}(\textsc{Cat}_2, \times)$$

It is again monoidal, with respect to cartesian product (we will need this).

The category of internal monoids

Suppose Ω has a terminal object $*_\Omega$. Then the unique morphisms $1_\Omega \to *_\Omega$ and $*_\Omega \otimes_\Omega *_\Omega \to *_\Omega$ give $*_\Omega$ the structure of a (strict) pseudomonoid in Ω.

Define the category of monoids internal to a pseudomonoid $\mathcal{C}$ in a monoidal 2-category $(\Omega, \otimes_\Omega)$ as

$$Mon_{\mathcal{C}}^{\Omega} = PsMon(\Omega, \otimes_\Omega)(*_\Omega, \mathcal{C}).$$

Therefore, a monoid internal to $\mathcal{C} \in PsMon(\Omega, \otimes_\Omega)$ is the data of

- a morphism $\bar{X} : *_\Omega \to \mathcal{C}$
- a 2-morphism

$$\begin{tikzcd} *_\Omega \arrow[r, "\bar{X}\otimes_\Omega\bar{X}"] \arrow[dr, "\bar{X}"'] & \mathcal{C} \otimes_\Omega \mathcal{C} \arrow[d, "\mu_{\mathcal{C}}"] \arrow[dl, phantom, "\Swarrow m_{\bar{X}}"] \\ & \mathcal{C} \end{tikzcd}$$

verifying associativity and unit conditions.

- $(\Omega, \otimes_\Omega) = (\text{CAT}, \times)$:
    - $Mon^{\Omega}_{\mathcal{C}}$ = category of monoids in a monoidal category $\mathcal{C}$, that is:
        - an object $x \in \mathcal{C}$
        - an associative morphism $x \otimes_{\mathcal{C}} x \to x$ in $\mathcal{C}$ an a unital morphism $1 \to x$
    - Let $(\mathcal{C}, \otimes_{\mathcal{C}})$ be a monoidal cocomplete category s.t. $\otimes_{\mathcal{C}}$ distributes over colimits. If $X, Y \in \mathcal{C}^{\mathbb{N}}$, then

$$X \circ Y(n) = \coprod_{n_1+\ldots+n_r=n} X(r) \otimes_{\mathcal{C}} \bigotimes_{i=1}^{r}{}_{\mathcal{C}} Y(n_i)$$

makes $(\mathcal{C}^{\mathbb{N}}, \circ)$ into a monoidal category. Its category of internal monoids is then the category of operads with values in $\mathcal{C}$:

$$Mon_{(\mathcal{C}^{\mathbb{N}},\circ)}^{(\text{Cat},\times)} \cong Op_{\mathcal{C}}.$$

- $(\Omega, \otimes_\Omega) = (\text{Cat}^{\mathbb{N}}, \circ)$:

  Let $\mathcal{P} = \{\mathcal{P}(r)\}$ be a pseudo-operad, then its category of internal monoids

  $$Mon_{(\mathcal{P},\mu_{\mathcal{P}})}^{(\text{Cat}^{\mathbb{N}},\circ)} \cong Op_{\mathcal{P}}$$

  gives rise to Batanin's notion of an operad internal to a categorical operad:

  - an object $X(r) \in \mathcal{P}(r)$ for all $r \in \mathbb{N}$
  - a morphism $\mu_{\mathcal{P}}(X(r), X(n_1), ..., X(n_r)) \to X(n)$ in $\mathcal{P}(n)$, ...

Extending Day convolution

- Day convolution for $\mathcal{V}$-enriched categories can be reformulated as follow:

$$[-,\mathcal{V}]_{\mathcal{V}} : \mathit{PsMon}(\mathrm{CAT}_{\mathcal{V}}, \times) \to \mathit{PsMon}(\mathrm{CAT}_{\mathcal{V}}, \times).$$

- We needed a 2-category Ω, with
    - a closed structure
    - Kan extensions
    - a monoidal structure $(\Omega, \otimes_{\Omega})$
    - pseudomonoids with respect to it $\mathit{PsMon}(\Omega, \otimes_{\Omega})$
    - an external product

Let $(\Omega, \otimes_\Omega)$ be a monoidal 2-category which is cartesian closed. Write

$$[-,-]_\Omega : \Omega^{op} \times \Omega \to \Omega$$

for the internal hom 2-functor. Now suppose this functor is a pseudomonoidal morphism in the monoidal 2-category $(\mathit{Mon}\text{CAT}_2, \times)$. This means we have morphisms

$$\bar{\otimes} : [\mathcal{C}_1, \mathcal{D}_1]_\Omega \otimes_\Omega [\mathcal{C}_2, \mathcal{D}_2]_\Omega \to [\mathcal{C}_1 \otimes_\Omega \mathcal{C}_2, \mathcal{D}_1 \otimes_\Omega \mathcal{D}_2]_\Omega$$

verifying some naturality, associativity (etc) conditions. We call it the **external product**.

Let $(\Omega, \otimes_\Omega)$ be a monoidal 2-category which is cartesian closed, such that the internal hom 2-functor is pseudomonoidal.

Let $(\mathcal{C}, \mu_{\mathcal{C}}), (\mathcal{V}, \mu_{\mathcal{V}}) \in PsMon(\Omega, \otimes_\Omega)$ be pseudomonoids.

We can finally generalize Day convolution:

$$\begin{tikzcd}
{[\mathcal{C}, \mathcal{V}]_\Omega \otimes_\Omega [\mathcal{C}, \mathcal{V}]_\Omega} \arrow[rr, "\otimes^\Omega_{Day}"] \arrow[d, "\bar{\otimes}"'] & & {[\mathcal{C}, \mathcal{V}]_\Omega} \\
{[\mathcal{C} \otimes_\Omega \mathcal{C}, \mathcal{V} \otimes_\Omega \mathcal{V}]_\Omega} \arrow[d, "{[id, \mu_{\mathcal{V}}]_\Omega}"'] & & \\
{[\mathcal{C} \otimes_\Omega \mathcal{C}, \mathcal{V}]_\Omega} \arrow[uurr, "Lan_{\mu_{\mathcal{C}}}"'] & &
\end{tikzcd} .$$

Theorem

The morphism thus defined

$$\otimes^{\Omega}_{Day} : [\mathcal{C}, \mathcal{V}]_{\Omega} \otimes_{\Omega} [\mathcal{C}, \mathcal{V}]_{\Omega} \longrightarrow [\mathcal{C}, \mathcal{V}]_{\Omega}$$

gives $([\mathcal{C}, \mathcal{V}]_{\Omega}, \otimes^{\Omega}_{Day})$ *the structure of a pseudomonoid in* $(\Omega, \otimes_{\Omega})$.

Moreover, this gives rise to a 2-functor

$$[-, \mathcal{V}]_{\Omega} : PsMon(\Omega, \otimes_{\Omega}) \to PsMon(\Omega, \otimes_{\Omega}).$$

Proposition The cartesian product defines a pseudomonoidal morphism

$$\times : \Omega \times \Omega \to \Omega.$$

This means we have morphisms

$$(\mathcal{C}_1 \times \mathcal{D}_1) \otimes_\Omega (\mathcal{C}_2 \times \mathcal{D}_2) \to (\mathcal{C}_1 \otimes_\Omega \mathcal{C}_2) \times (\mathcal{D}_1 \otimes_\Omega \mathcal{D}_2).$$

Therefore, it takes pseudomonoids to pseudomonoids:

$$\times : PsMon\left(\Omega \times \Omega, \otimes_{\Omega\times\Omega}\right) \to PsMon\left(\Omega, \otimes_\Omega\right).$$

Moreover, we have

$$PsMon\left(\Omega \times \Omega, \otimes_{\Omega\times\Omega}\right) \cong PsMon\left(\Omega, \otimes_\Omega\right) \times PsMon\left(\Omega, \otimes_\Omega\right)$$

Consequence The 2-category $PsMon(\Omega, \otimes_\Omega)$ is monoidal.

Remark This monoidal structure is actually the cartesian one.

Conclusion

We can generalize Day convolution in any cartesian closed monoidal 2-category $(\Omega, \otimes_\Omega)$ such that the internal hom is a morphism of pseudomonoids, whenever the internal left Kan extension exists along the morphisms we want. In a very synthetic way:

Day convolution just says $\mathit{PsMon}(\Omega, \otimes_\Omega)$ is cartesian closed.

$$- \times \mathcal{V} : \mathit{PsMon}\,(\Omega, \otimes_\Omega) \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \mathit{PsMon}\,(\Omega, \otimes_\Omega) : \left([-, \mathcal{V}]_\Omega\,, \otimes^{\Omega}_{\mathit{Day}}\right)$$

An example

Take $\Omega = \left(\mathrm{CAT}^{\mathbb{N}}, \circ, \times, [-,-]_{\mathbb{N}}\right)$, with

$$[\mathcal{P}, \mathcal{Q}]_{\mathbb{N}}(n) = [\mathcal{P}(n), \mathcal{Q}(n)].$$

Let $\bar{Sets} \in \mathrm{CAT}^{\mathbb{N}}$ be the constant sequence of categories $\bar{Sets}(n) = Sets$. The cartesian product makes $\bar{Sets}$ into a pseudomonoid in $(\mathrm{CAT}^{\mathbb{N}}, \circ)$:

$$\bar{Sets} \circ \bar{Sets}(n) = \coprod_{n_1+\ldots+n_r=n} \bar{Sets}(r) \times \prod_{i=1}^{r} \bar{Sets}(n_i) \to \bar{Sets}(n)$$

given by

$$Sets \times \prod_{i=1}^{r} Sets \to Sets.$$

Let $\mathcal{P} \in PsMon\left(\mathrm{CAT}^{\mathbb{N}}, \circ\right)$ be a pseudo operad, thanks to generalized Day convolution product, the sequence of presheaves

$$[\mathcal{P}, Sets]_{\mathbb{N}}(n) = [\mathcal{P}(n), Sets]$$

has the structure of a pseudo operad.

Take $\mathcal{P} = \Upsilon$ the operad whose arity r operations are given by the category $\Upsilon(r)$ whose

- objects are trees with r leaves
- morphisms are edge contractions

with composition given by grafting of trees :

$$\Upsilon(r) \times \prod_{i=1}^{r} \Upsilon(n_i) \to \Upsilon(n).$$

We have an explicit description of day convolution:

$$[\Upsilon^{op}, Sets]_{\mathbb{N}} \circ [\Upsilon^{op}, Sets]_{\mathbb{N}} \to [\Upsilon^{op}, Sets]_{\mathbb{N}}$$

induced by

$$[\Upsilon^{op}(r), Sets] \times \prod_{i=1}^{r}[\Upsilon^{op}(n_i), Sets] \to [\Upsilon^{op}(n), Sets],$$

$$(X, X_1, ..., X_r) \mapsto X(X_1, ..., X_r)$$

where

$$\begin{array}{rccc} X(X_1, ..., X_r): & \Upsilon^{op}(n) & \to & Sets \\ & T & \mapsto & \displaystyle\coprod_{T_0(T_1,\ldots,T_r)=T} X_{T_0} \times \prod_{i=1}^{r} X_{i\,T_i}. \end{array}$$

Therefore, operads internal to $[\Upsilon^{op}, Sets]_{\mathbb{N}}$ = sequence of presheaves

$$X(n) : \Upsilon^{op}(n) \to Sets$$

together with an associative composition

$$X(r)_T \times \prod_{i=1}^{r} X(n_i)_{T_i} \to X(n)_{T(T_1,\ldots,T_r)}.$$